

Architectural Evolution of nopCommerce

Final Presentation

Scenario C - Omnichannel Commerce Core

TP1 Group 02

Presented by:

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THE PROBLEM

The **nopCommerce** application must stay useful when an external Warehouse Management System or Point of Sale System lag or fail.

Objective is coordination under delay and degradation.

To do this we divided the work in three iterations

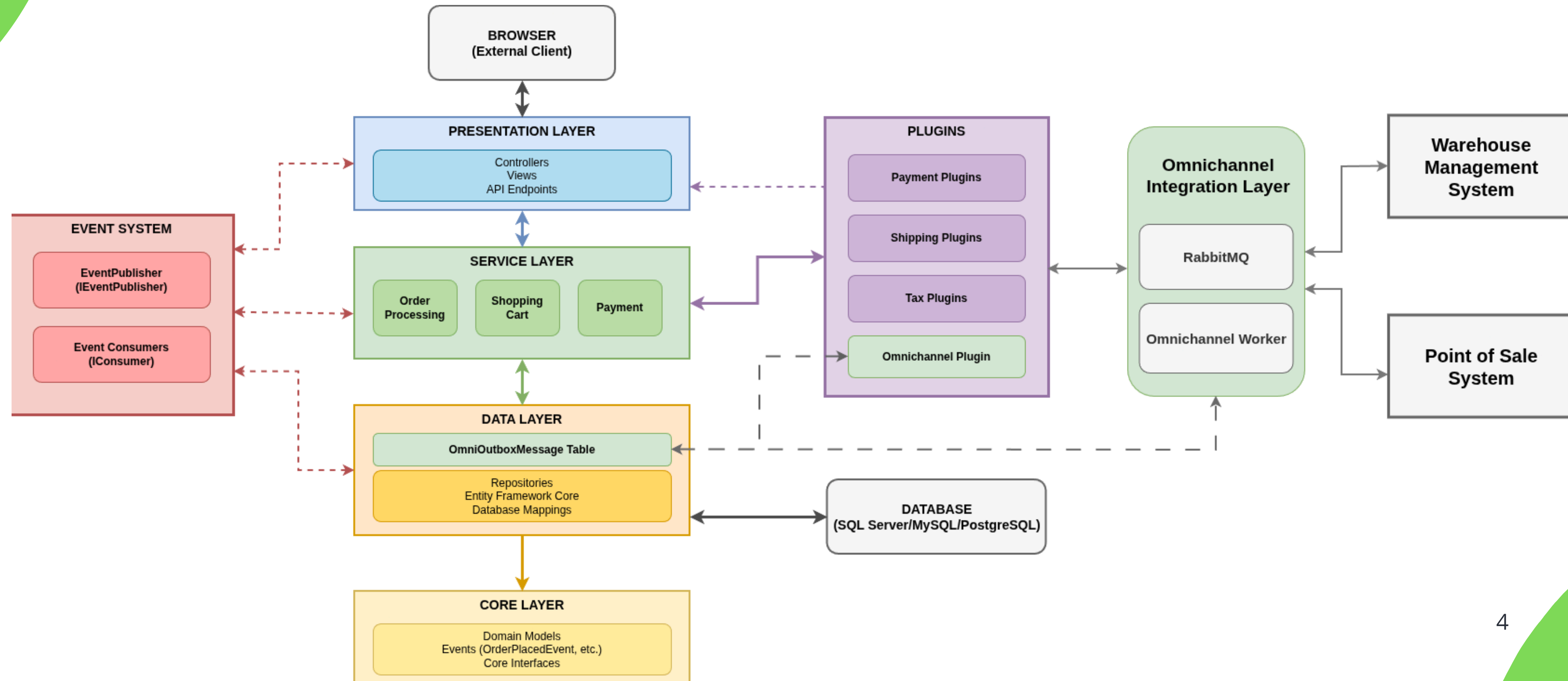
- **Resilience** - checkout still succeeds when the WMS is slow or down
- **Consistency** - duplicate or out-of-date events never corrupt state
- **Traceability** - any order is explainable end-to-end from the admin view

THE APPROACH

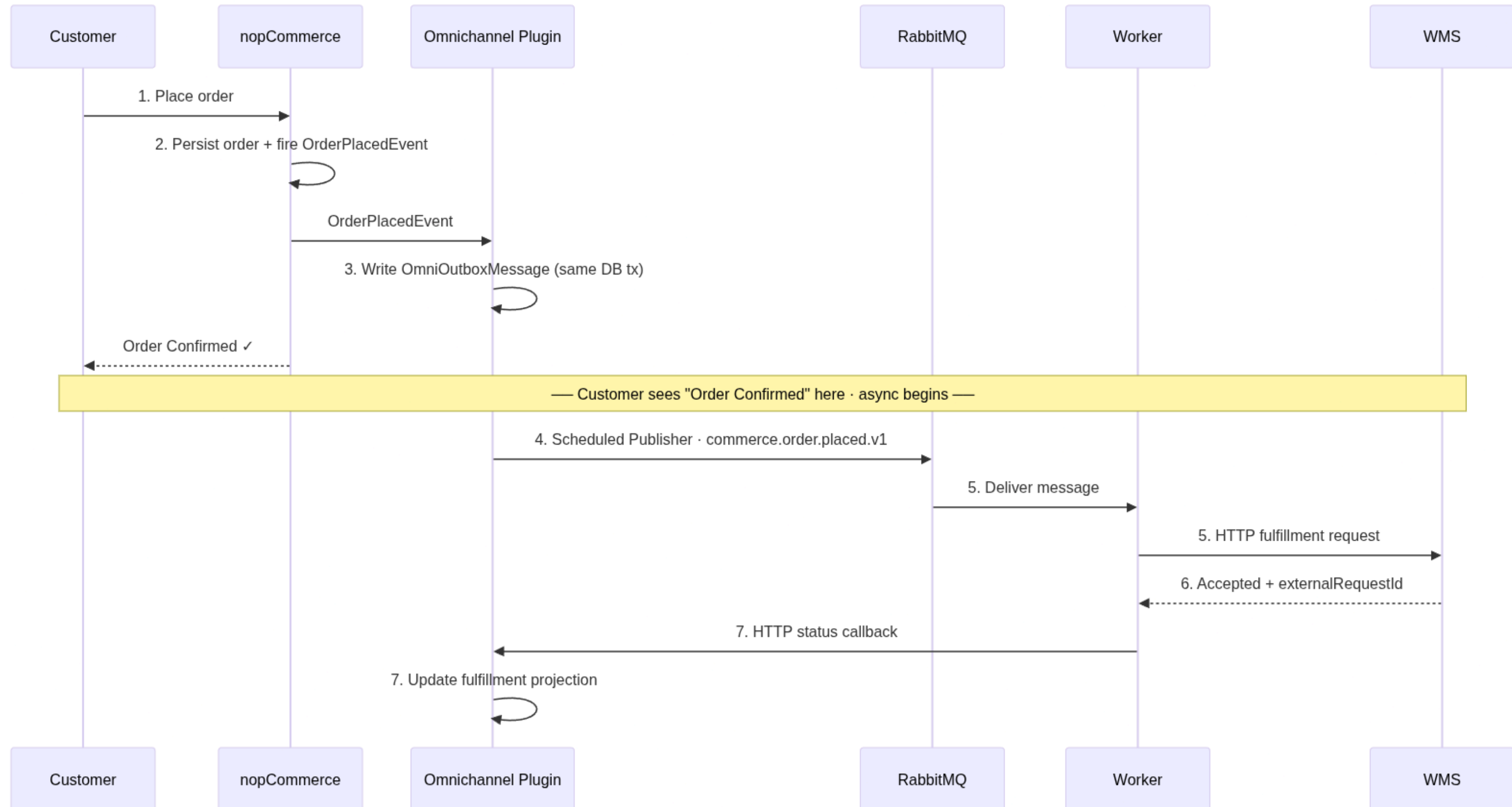
Framework chosen was **Attribute Driven Design (ADD)** with borrowed pieces from ACDM and ADM.

Driver	QA	Description	Iteration
Resilience	01	checkout survives a WMS outage and self-recovers	1
Performance	02	integration never blocks the checkout thread	1
Consistency	03	duplicates or stale events never corrupt state	2
Traceability	04	any order is explainable end-to-end from admin	3
Opearability	05	queue, retries & DLQ visible in one dashboard	3

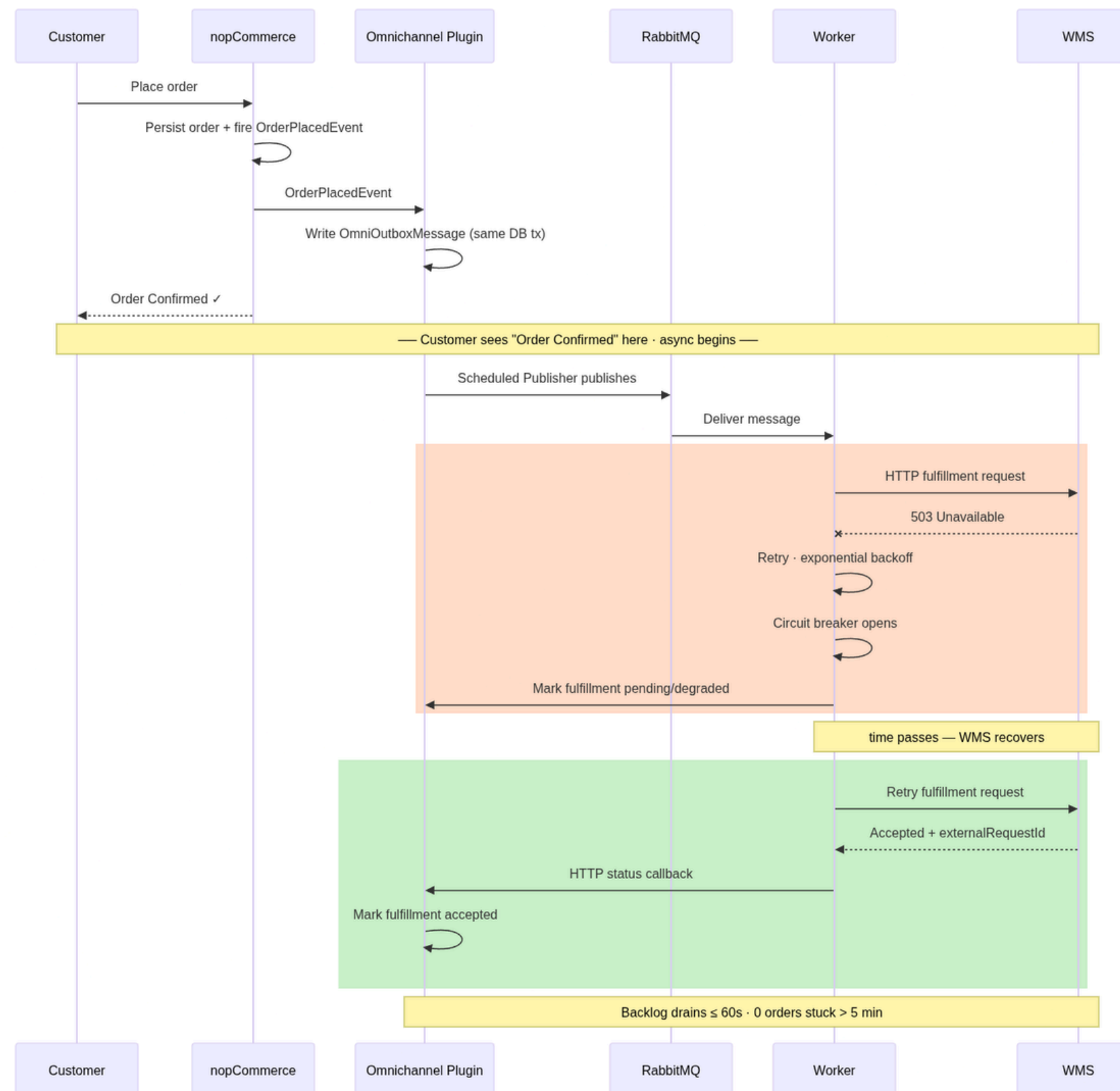
THE ARCHITECTURE

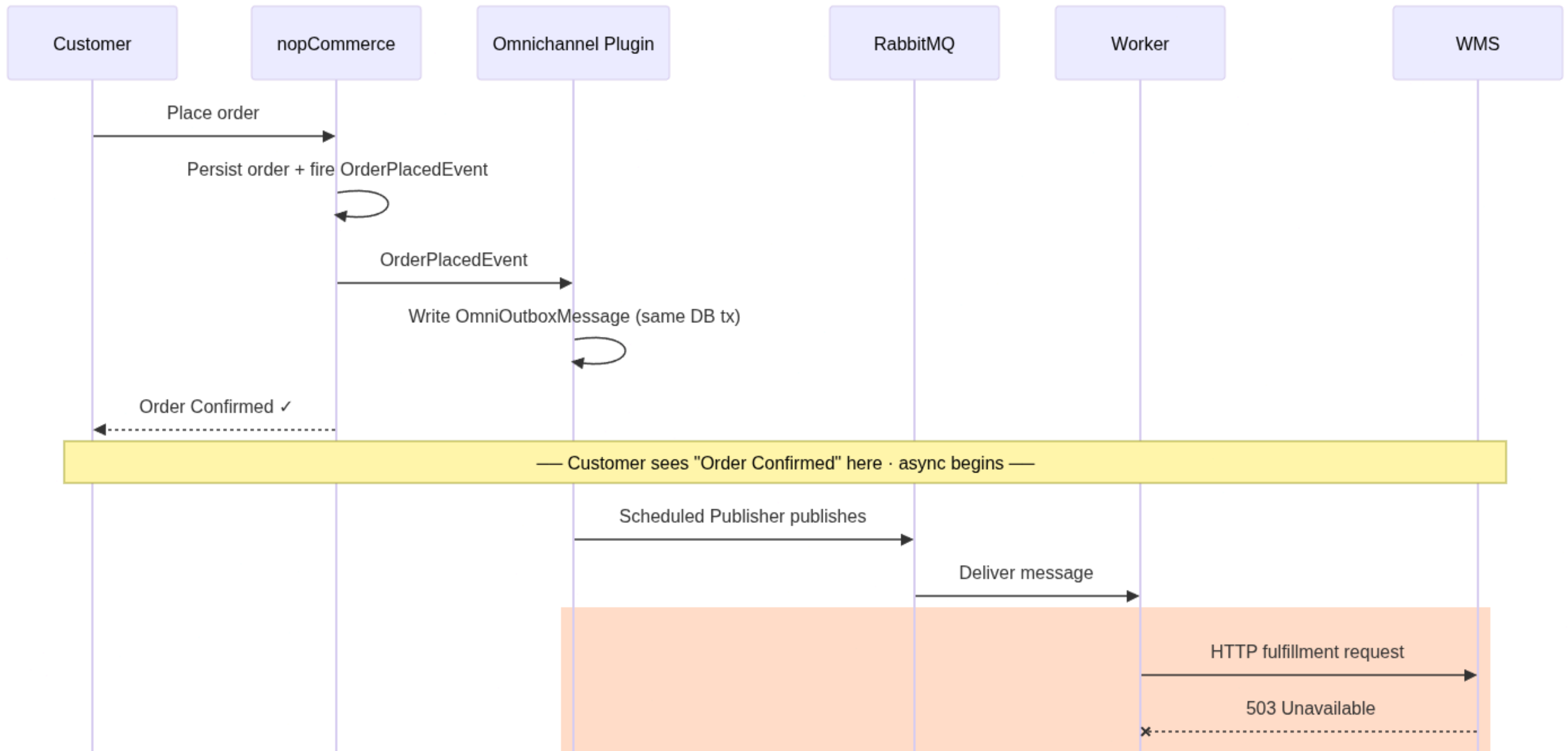


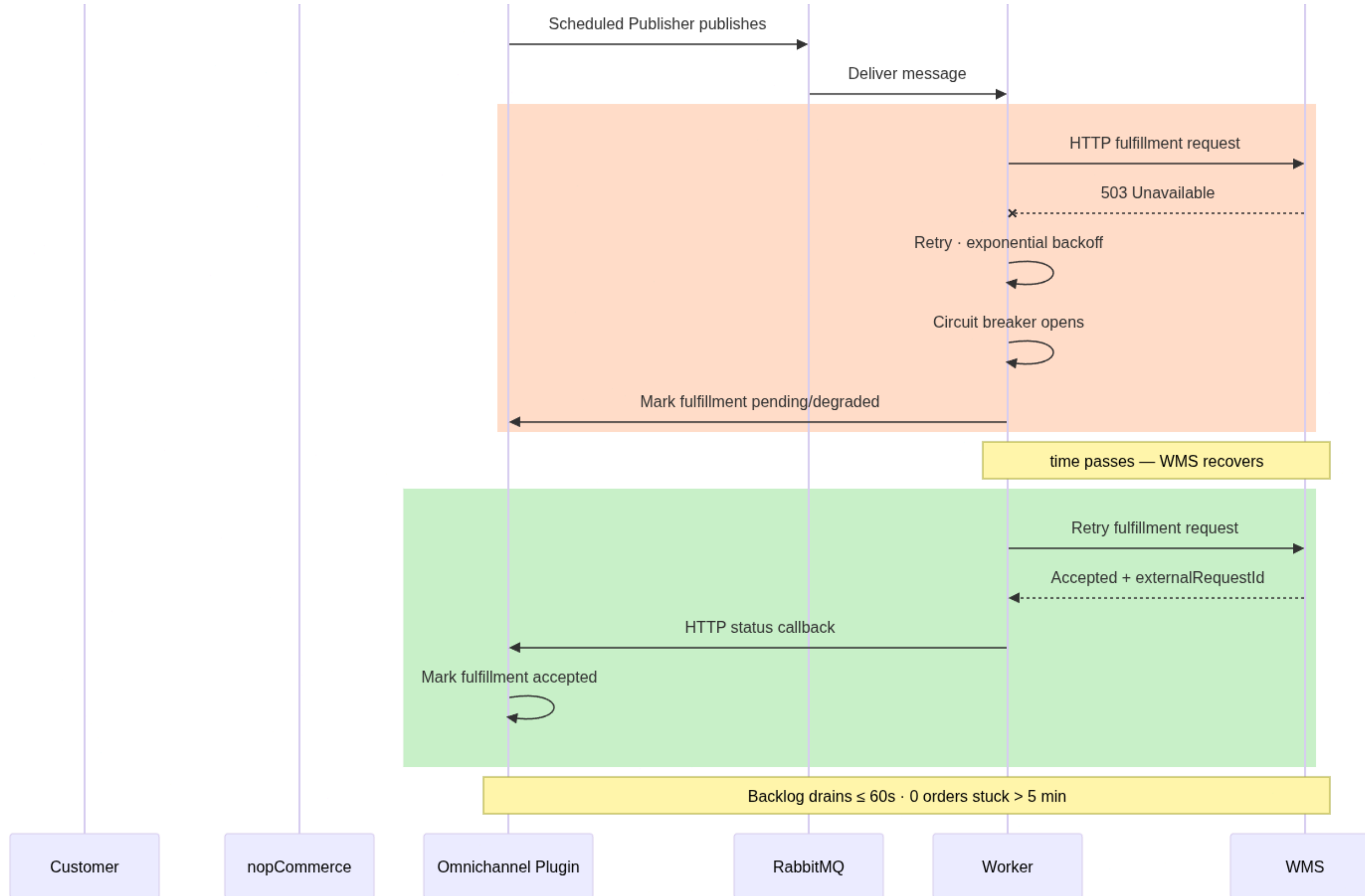
THE ORDER FLOW



THE ORDER FLOW







IMPORTANT DECISIONS

Decision	Tactic	Consequences	ADR
Transactional Outbox + Reconsiler	crash-safe event capture	order can't exist with no integration trail	0003 + 0011
Idempotent Inbox	<i>messageId</i> dedup + <i>sourceVersion</i> guard	at-least-once delivery can't duplicate or apply stale data	0006
Projection-first Stock	shadow table, write through deferred	stale POS event can't corrupt core stock	0007
Retry + Circuit Breaker + DLQ	guards the worker's WMS call	WMS degradation contained, poison isolated	0003

LIVE DEMO

EVIDENCE

QA 01 - Resilience

Target	Results
Degraded checkout P95 $\leq$ 2085 ms	1822 ms
Checkout failures from WMS = 0	50/50 placed · 0 failures
Backlog drain $\leq$ 60 s	8 s
0 orders pending > 5 min	0 pending

EVIDENCE

QA 02 - Performance

Target	Results
outbox write $\leq$ 100 ms	avg 10.3 ms · max 59 ms
plugin cost on order-placement call	366 → 405 ms (~10 ms is the outbox write)

EVIDENCE

QA 03 - Consistency

Target	Results
Duplicate rejected ≤ 50 ms	17.5 ms
Same event delivered twice $\rightarrow$ handled once	1 inbox row (not 2) · 0 order rows
Out-of-order (stale) update ignored	v44 after v45 $\rightarrow$ rejected stock stays at v45 value

EVIDENCE

QA 03 - Consistency

Outbox (commerce.order.placed.v1)						
Id	MessageId	Status	Retries	Created (UTC)	Published (UTC)	Last error
10	b91dd07b-d7f2-438b-9c60-5ac9fb6532f3	Published	0	2026-06-02 13:27:42Z	2026-06-02 13:27:43Z	
Fulfillment projection (OmniOrderFulfillment)						
Status				Accepted		
ExternalRequestId				WMS-REQ-15		
TrackingNumber				—		
Reason				—		
Inbox (idempotency ledger)						
Id	MessageId	EventType		Status	Received (UTC)	
17	2861e4e0-d9e6-420b-aa1a-e98d73ca84fa	fulfillment.status.changed.v1		Processed	2026-06-02 13:27:44Z	

EVIDENCE

QA 04 - Traceability

Projection	Purpose	Rows	Rows
OmniOutboxMessage	Durable outbound events waiting for publication.	5	5
OmniInboxMessage	Inbound message ledger for idempotency and duplicate detection.	0	5
OmniOrderFulfillment	Order fulfillment projection owned by the plugin.	0	5
↳ pending / degraded	Fulfillment rows still awaiting or degraded — operability signal (QA-4).	5	0
OmniStockSyncState	Cross-channel stock projection with source version tracking.	0	0

EVIDENCE

QA 05 - Opearability

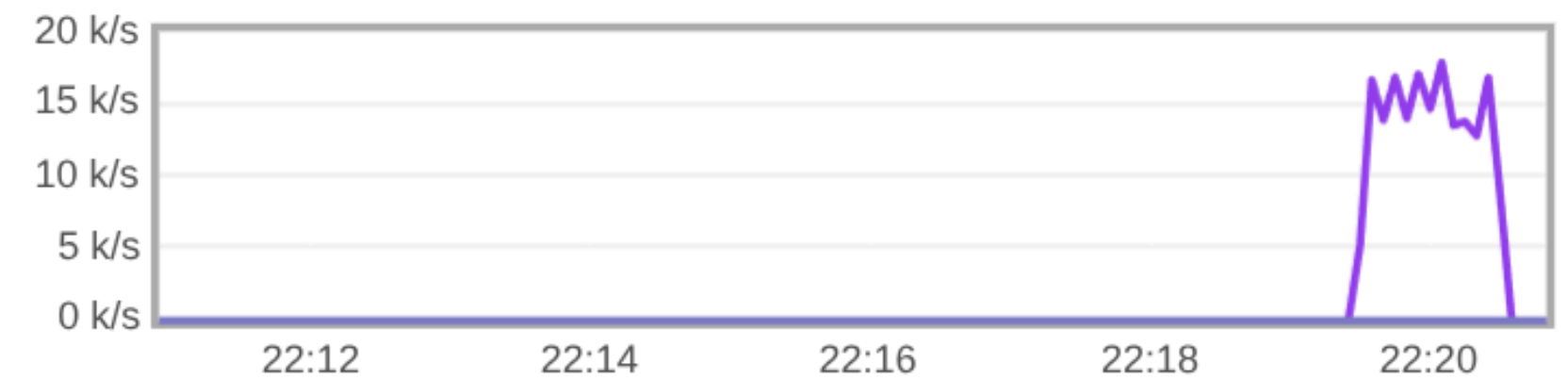
Queue wms.order.placed

▼ Overview

Queued messages last ten minutes ?



Message rates last ten minutes ?



FUTURE WORK

Today	Next step
Projection can drift from core stock	Stock write-through once drift is measured
App-level <i>messageld</i> dedup	DB unique constraint on <i>messageld</i>
Inbox / outbox uncapped	Retention + capping policy
WMS / POS are simulators	Swap in real systems via the versioned contracts
Demo-token auth on callbacks	Real auth (signed tokens / mTLS)



deti

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